The Journal of Open Source Software

pyvisco: A Python library for identifying Prony series parameters of linear viscoelastic materials

Martin Springer [1¶]

1 National Laboratory of the Rockies, United States ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Kevin M. Moerman

Reviewers:

- @Noah-Parsons
- @jorge-ramirez-upm

Submitted: 04 June 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

Polymeric materials commonly used in engineering applications, such as the encapsulants and backsheets in photovoltaic modules, adhesives in structural joints, and rubbers in seals and dampers, exhibit time- and temperature-dependent mechanical behavior. The theory of linear viscoelasticity provides a framework to describe this behavior, and the Generalized Maxwell model represented through a Prony series is widely adopted in commercial finite element analysis (FEA) software to simulate viscoelastic response of thermorheologically simple materials under mechanical and thermal loads. Identifying the Prony series parameters from experimental measurements is, however, a non-trivial multi-step process that involves data conditioning, time–temperature superposition (TTSP), shift function fitting, and constrained non-linear least-squares optimization.

`pyvisco` is an open-source Python library that automates this workflow. Given either raw isothermal measurements at multiple temperatures (from dynamic mechanical thermal analysis or relaxation experiments) or a pre-assembled master curve, the library constructs the master curve via TTSP, fits Williams–Landel–Ferry (WLF) and polynomial shift functions, and identifies the relaxation times τ_i and moduli E_i (or shear moduli G_i) of a Prony series compatible with the Generalized Maxwell model. An optional minimization routine produces a reduced-term Prony series suitable for computationally efficient FEA simulations. The library is accessible through a programmatic application programming interface (API) and through a Jupyter-based graphical user interface (`LinViscoFit.ipynb`) that runs without local installation as a Voila application on Hugging Face Spaces. Library outputs are verified against the curve-fitting routine of ANSYS APDL 2021 R1 (ANSYS Inc., Canonsburg, Pennsylvania) using bundled reference cases.

Statement of need

Linear viscoelastic material models are routinely required by researchers and engineers working with polymers in structural, thermal, and reliability analyses. The underlying mathematical framework is well established and combines the Boltzmann superposition integral, the Generalized Maxwell model, time–frequency interconversion via Fourier transformation, and TTSP (Brinson & Brinson, 2015; Ferry, 1980; Park & Schapery, 1999; Roylance, 2001; Williams et al., 1955). This framework forms the basis of the characterization workflow described by Springer & Bosco (2020), on which `pyvisco` is built. However, translating raw experimental data into a Prony series suitable for FEA is tedious and error-prone. The process typically requires:

- shifting isothermal measurement sets in the frequency or time domain to form a master curve at a reference temperature;
- fitting a shift function (e.g., WLF (Williams et al., 1955), Arrhenius, or low-degree polynomial) to parameterize the temperature dependence;

DRAFT

- selecting an appropriate number and distribution of Prony terms;
- solving a bound-constrained least-squares problem to identify τ_i and E_i while enforcing $E_\infty \geq 0$ and $\sum \alpha_i \leq 1$;
- optionally reducing the number of Prony terms to keep FEA simulations tractable.

Many researchers implement this pipeline ad hoc in spreadsheets or short scripts. Furthermore, commercial dynamic mechanical thermal analysis (DMTA) software (e.g., Netzsch GABO Eplexor, TA Instruments TRIOS) typically allows only a subset of these steps and offers limited interoperability with FEA pre-processors. `pyvisco` was designed to fill this gap with a reproducible, open-source workflow that:

1. accepts both raw and pre-shifted master-curve inputs in CSV or Eplexor Excel format;
2. produces Prony series output cards directly usable in commercial FEA codes;
3. is accessible to non-programmers through a browser-based notebook interface (powered by `ipywidgets`) without local installation; and
4. is verified against ANSYS APDL on a reference benchmark set kept in the repository.

The library targets researchers characterizing polymeric materials (e.g., photovoltaic (PV) encapsulants, adhesives, elastomers, hydrogels) and engineers preparing material cards for viscoelastic FEA simulations.

State of the field

A number of tools touch parts of the viscoelastic characterization workflow:

- **Commercial DMTA software** (Netzsch GABO Eplexor, TA Instruments TRIOS, Anton Paar RheoCompass) can construct master curves and fit shift functions, but Prony series export and reduced-term optimization for FEA are limited or absent, and licensing restricts reproducibility.
- **pyRheo** (Miranda-Valdez, Coffeng, et al., 2024) focuses on rheological model fitting (fractional and integer-order constitutive models) but does not target Prony-series identification for the Generalized Maxwell model used in FEA, nor does it integrate the full TTSP pipeline.
- **RepTate** (Boudara et al., 2020) is an open-source rheology data analysis suite with a Qt GUI; while powerful for rheological model selection, it is primarily a desktop application and does not export Prony-series cards specifically tuned for finite element use.
- **Bespoke MATLAB/Python scripts** are common in the literature (Barrientos et al., 2019; Park & Schapery, 1999) but are typically not packaged, tested, or maintained beyond a single publication.

`pyvisco` was built as a focused, reusable package because none of the above combine: (a) a full raw-data-to-Prony-series workflow in a single tool, (b) an optimization routine that yields a reduced-term Prony series suitable for FEA, (c) verification against a commercial FEA software's own fitting routine, and (d) a zero-install browser interface that lowers the barrier for non-Python users working in materials labs. Where alternatives are well suited to their target use cases, `pyvisco` complements them by closing the loop between DMTA measurements and FEA input decks.

Software design

`pyvisco` is organized as a small set of single-purpose modules (`load`, `master`, `shift`, `prony`, `opt`, `out`, `verify`, `styles`, `inter`) that each expose pure functions operating on `pandas.DataFrame` inputs. This functional, dataframe-centric design was chosen so that:

- each step (loading, master-curve assembly, shift-function fitting, Prony fitting, optimization, output) can be invoked, inspected, and unit tested in isolation;

DRAFT

- intermediate states are first-class data artifacts that can be cached, exported, or visualized without re-running upstream computations; and
- the user interface layer (`inter.py`, an `ipywidgets`-based controller) is a thin orchestration layer with no domain logic of its own.

The Prony fit is formulated as a bound-constrained non-linear least-squares problem solved with SciPy's L-BFGS-B implementation (Virtanen et al., 2020), following the formulation of Barrientos et al. (2019) and the interconversion methodology of Park & Schapery (1999). Time-domain to frequency-domain conversion of relaxation data uses the Fourier transformation summarized in Springer & Bosco (2020). Master curves are optionally smoothed using a Savitzky–Golay filter (Savitzky & Golay, 1964) before Prony-series fitting to suppress measurement noise without broadening the relaxation spectrum. The reduced-term optimization routine progressively removes Prony terms and refits, using the ratio of residuals ($R^2_{\text{opt}} \approx 1.5\, R^2_0$) as a default stopping heuristic that the user can override. Numerical routines build on `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `pandas` (McKinney, 2010), and `matplotlib` (Hunter, 2007); the graphical user interface (GUI) is served by Voila as a Docker-based Hugging Face Space.

Outputs include the fitted Prony parameters as CSV, plots in PNG, and a single ZIP archive bundling all results for downstream FEA use. The `verification/` directory contains end-to-end notebooks comparing `pyvisco` fits against ANSYS APDL 2021 R1 reference cases; these notebooks are executed on every push via GitHub Actions to guard against regressions.

Research impact statement

`pyvisco` originated to support viscoelastic characterization of PV module encapsulants and interconnect adhesives, as part of the reliability and durability research described by Springer & Bosco (2020). Since its initial public release in 2022 the package has been published on PyPI, archived on Zenodo with a DOI per release (Springer, 2022), indexed by Read the Docs, and developed openly on GitHub with continuous integration, an issue tracker, a tagged release history, and a public changelog spanning the v1.x and v2.x series.

The v2.0 release re-organized the codebase into the modular layout described above, added a verified test suite, modernized the packaging (PEP 621 `pyproject.toml`, Python 3.11), and introduced standalone tutorial scripts in addition to the interactive notebook so the library can be used in headless and continuous-integration contexts.

Beyond its originating use case, `pyvisco` has been adopted by independent research groups across a range of polymer-mechanics domains. To date the software has been used in at least eight peer-reviewed publications and technical reports, including: fractional-calculus modeling of methylcellulose aqueous systems (Miranda-Valdez, Puente-Córdova, et al., 2024); finite-strain viscoelastic-damage modeling of solid propellants (Gouhier et al., 2024); characterization and modeling of waterborne epoxy varnish coatings for steel/epoxy laminates in electric machines (Tiefenthaler et al., 2025); structural-dynamic characterization of polymeric foams via time–temperature superposition (Le Barbenchon et al., 2025); fatigue testing of bitumen binders using column specimens (Mangalath Shine et al., 2023); nano-indentation DMTA characterization of polymeric materials in microelectronic packaging (Lin et al., 2024); viscoelastic characterization of polymer films (Lin & Gamez, 2025); and biomechanical modeling of handball-induced head injuries (Johansson & Nilsson, 2025). These applications span food science, aerospace propellants, electric-machine manufacturing, cellular polymers, road-pavement engineering, microelectronics packaging, and sports biomechanics, demonstrating that the workflow generalizes well beyond its original PV context.

By providing a reproducible, open, and FEA-oriented Prony identification workflow, `pyvisco` complements existing rheology packages and lowers the barrier for materials researchers to share viscoelastic material cards in a form directly usable by the wider engineering community.

DRAFT

AI usage disclosure

Generative AI tools were used during the preparation of the v2.x release and of this manuscript:

- **Tools/models used:** GitHub Copilot in Visual Studio Code (Claude Opus 4.7 and GPT-class models accessed through Copilot Chat), used during development between 2024 and 2026.
- **Where used:** (a) refactoring of the v1.x codebase into the modular layout, including renaming and reorganization of functions; (b) drafting and updating user-facing documentation (`README.md`, the changelog under `docs/changelog/`, docstrings, tutorial scripts); (c) drafting unit-test scaffolding under `tests/`; (d) initial drafting of this `paper.md`.
- **Where not used:** the core numerical formulation of the Prony fit, shift-function fits, and time-temperature superposition algorithms predate the use of AI assistance and were authored by the human author; the verification notebooks against ANSYS APDL and their reference data were authored manually.
- **Verification:** All AI-assisted code was reviewed by the author before commit, executed locally, and validated by the existing test suite and the end-to-end verification notebooks comparing `pyvisco` outputs against ANSYS APDL. AI-assisted prose was reviewed and edited by the author for accuracy. The author takes full responsibility for the correctness, originality, and licensing of all materials in this submission.

Acknowledgements

The author thanks colleagues for early feedback on the workflow, and the maintainers of `numpy`, `scipy`, `pandas`, `matplotlib`, and `ipywidgets` on which `pyvisco` builds.

This work was authored by the National Laboratory of the Rockies for the U.S. Department of Energy (DOE), operated under Contract No. DE-AC36-08GO28308. Funding was provided as part of the Durable Module Materials Consortium 2 (DuraMAT 2), funded by the U.S. Department of Energy Office of Critical Minerals and Energy Innovation Integrated Energy Systems Office, agreement number 38259. The views expressed in the article do not necessarily represent the views of the DOE or the U.S. Government. The U.S. Government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

References

Barrientos, E., Pelayo, F., Noriega, Á., Lamela, M. J., Fernández-Canteli, A., & Tanaka, E. (2019). Optimal discrete-time Prony series fitting method for viscoelastic materials. *Mechanics of Time-Dependent Materials*, *23*(2), 193–206. https://doi.org/10.1007/s11043-018-9394-z

Boudara, V. A. H., Read, D. J., & Ramirez, J. (2020). RepTate rheology software: Toolkit for the analysis of theories and experiments. *Journal of Rheology*, *64*(3), 709–722. https://doi.org/10.1122/8.0000002

Brinson, H. F., & Brinson, L. C. (2015). *Polymer engineering science and viscoelasticity: An introduction* (2nd ed.). Springer. https://doi.org/10.1007/978-1-4899-7485-3

Ferry, J. D. (1980). *Viscoelastic properties of polymers* (3rd ed.). John Wiley & Sons. ISBN: 978-0-471-04894-7

Gouhier, F., Diani, J., & Vandenbroucke, A. (2024). A finite strain viscoelastic model with damage and tension–compression asymmetry considerations for solid propellants. *Mechanics*

DRAFT

of Materials, *199*, 105152. https://doi.org/10.1016/j.mechmat.2024.105152

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Johansson, A., & Nilsson, C. (2025). *Modeling handball-induced head injuries* [Master's thesis, Chalmers University of Technology]. https://odr.chalmers.se/bitstreams/7fb05d10-493e-40f1-8c56-ccc7a0ea0fdd/download

Le Barbenchon, L., Girardot, J., & Kopp, J.-B. (2025). Investigation of structural dynamic effects on the mechanical behavior of a polymeric foam by means of time–temperature superposition. *Journal of Dynamic Behavior of Materials*. https://doi.org/10.1007/s40870-025-00504-7

Lin, H.-Y., & Gamez, A. (2025). A novel methodology for characterizing and validating viscoelastic and thermal expansion properties of polymer films. *2025 IEEE 75th Electronic Components and Technology Conference (ECTC)*. https://doi.org/10.1109/ECTC51687.2025.00264

Lin, H.-Y., Gurrum, S., Gamez, A., & others. (2024). Novel nano-indentation dynamic mechanical thermal analysis (NiDMTA) technique for characterizing viscoelastic properties of polymeric materials in microelectronic packaging. *2024 23rd IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm)*. https://doi.org/10.1109/ITherm55375.2024.10709446

Mangalath Shine, A., Cañon Falla, G., Leischner, S., & Wellner, F. (2023). Fatigue testing on bitumen binder using different column specimen shapes. *Road Materials and Pavement Design*. https://doi.org/10.1080/14680629.2023.2181067

McKinney, W. (2010). Data structures for statistical computing in Python. In S. van der Walt & J. Millman (Eds.), *Proceedings of the 9th python in science conference* (pp. 56–61). https://doi.org/10.25080/Majora-92bf1922-00a

Miranda-Valdez, I. Y., Coffeng, T., Zhou, Y., Viitanen, L., Hu, X., Jannuzzi, L. A. E., Puisto, A., Koivisto, J., & Alava, M. J. (2024). pyRheo: An open-source Python package for complex rheological data analysis. *Applied Rheology*. https://doi.org/10.48550/arXiv.2409.15466

Miranda-Valdez, I. Y., Puente-Córdova, J. G., Rendón-Hernández, F. Y., Aranda-Valdés, J. I., Coffeng, T., Viitanen, L., Koivisto, J., & Alava, M. J. (2024). Viscoelastic phenomena in methylcellulose aqueous systems: Application of fractional calculus. *Food Hydrocolloids*, *147*, 109334. https://doi.org/10.1016/j.foodhyd.2023.109334

Park, S. W., & Schapery, R. A. (1999). Methods of interconversion between linear viscoelastic material functions. Part I—a numerical method based on Prony series. *International Journal of Solids and Structures*, *36*(11), 1653–1675. https://doi.org/10.1016/S0020-7683(98)00055-9

Roylance, D. (2001). *Engineering viscoelasticity*. Department of Materials Science; Engineering, Massachusetts Institute of Technology.

Savitzky, A., & Golay, M. J. E. (1964). Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, *36*(8), 1627–1639. https://doi.org/10.1021/ac60214a047

Springer, M. (2022). *pyvisco: A Python library for identifying Prony series parameters of linear viscoelastic materials*. Zenodo. https://doi.org/10.5281/zenodo.6384954

DRAFT

Springer, M., & Bosco, N. (2020). Linear viscoelastic characterization of electrically conductive adhesives used as interconnect in photovoltaic modules. *Progress in Photovoltaics: Research and Applications*, *28*(7), 659–681. https://doi.org/10.1002/pip.3257

Tiefenthaler, M., Wallner, G. M., Marchfelder, C., Hild, S., Stadlbauer, W., & Lang, R. W. (2025). Simulation of steel/epoxy laminates for electric engines or generators: Part 1—Characterization and modeling of coating films based on waterborne epoxy varnishes. *Journal of Materials Science*. https://doi.org/10.1007/s10853-025-11041-y

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Williams, M. L., Landel, R. F., & Ferry, J. D. (1955). The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids. *Journal of the American Chemical Society*, *77*(14), 3701–3707. https://doi.org/10.1021/ja01619a008

DRAFT